

ChatMinds — Multi-AI Chat Studio

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A unified, responsive chat interface for multiple AI providers
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ChatMinds is a modern React + Vite chat application that brings together multiple AI providers (OpenAI, Google Gemini, Anthropic Claude, xAI Grok, DeepSeek, Groq, and LM Studio) into a single, beautiful interface. Perfect for experiments, local demos, and everyday AI conversations.

🔥 Features

🗂️ Multi-AI Provider Support

- **OpenAI** (GPT-4o, GPT-4o-mini)
- **Google Gemini** (Gemini 2.0 Flash, Gemini 1.5 Pro/Flash)
- **Anthropic Claude** (Claude 3.5 Sonnet, Claude 3 Opus/Haiku)
- **xAI Grok** (Grok-2, Grok-3)
- **DeepSeek** (DeepSeek Chat via OpenRouter)
- **Groq** (Llama, Mixtral models)
- **LM Studio** (Local models - connect to your local LM Studio server)

💬 Chat Features

- **Streaming Responses** — Real-time token-by-token response rendering
- **Multi-Chat Sessions** — Create, switch, and manage multiple conversations
- **Auto-Titled Chats** — Conversations auto-titled from first message
- **Persistent History** — All chats saved to localStorage
- **Global System Instruction** — Set AI personality/behavior across all chats
- **Markdown Rendering** — Beautiful formatting for AI responses
- **Copy Button** — One-click copy for any message

⚙️ Modern UI/UX

- **Responsive Design** — Optimized for desktop, tablet, and mobile
- **Dark/Light Theme** — Toggle with persistent preference
- **Mobile Hamburger Menu** — Sidebar with chat history, theme toggle, system instruction, and logout
- **Centered Branding** — Beautiful gradient ChatMinds logo on mobile
- **Clean, Minimal Interface** — Focus on the conversation

🗣️ Voice Features

- **Voice Input (Speech-to-Text)** — Dictate messages using your microphone
- **Voice Output (Text-to-Speech)** — Listen to AI responses with quality voice selection
- **Smart Voice Selection** — Prioritizes Google/Microsoft voices for clarity

📁 Multimodal Support

- **Image Upload** — Send images with text prompts
- **Camera Capture** — Take photos directly on mobile
- **Image Context Memory** — AI remembers previous image analysis

📄 Export & Sharing

- **PDF Export** — Download conversation as formatted PDF
- **Copy Messages** — Quick copy for individual messages

🔑 Authentication

- **Demo Auth System** — localStorage-based login/registration
- **Protected Routes** — Home redirects to login if unauthenticated

- **Easy to Upgrade** — Replace with Firebase/Auth0 for production

Mobile Experience

ChatMinds is designed mobile-first with:

- **Hamburger Sidebar** — Access chat history, theme, system instruction, and logout
- **Centered Header** — ChatMinds logo and name prominently displayed
- **Compact Assistant Selector** — Model picker positioned on the right
- **Touch-Friendly Controls** — Large tap targets for all buttons
- **Responsive Input** — Full-width text area with action buttons

Quick Start

1. Install Dependencies

```
cd chatgen
npm install
```

2. Configure Environment

Copy `.env.example` to `.env` and add your API keys:

```
VITE_OPEN_AI_API_KEY=sk-...
VITE_GOOGLE_AI_API_KEY=...
VITE_ANTHROPIC_AI_API_KEY=...
VITE_X_AI_API_KEY=...
VITE_DEEPSEEK_AI_API_KEY=...
VITE_GROQ_API_KEY=...
```

Note: Only configure the providers you want to use. LM Studio works locally without an API key.

3. Start Development Server

```
npm run dev
```

Open the URL printed by Vite (usually `http://localhost:5173`).

4. Access on Mobile (Same Network)

Find your computer's IP address and open `http://<your-ip>:5173` on your phone.

Project Structure

```

chatgen/
├── public/                                # Static assets
│   └── chat-bot.png                      # App logo
├── src/
│   ├── assistants/                      # AI provider adapters
│   │   ├── openai.js                   # OpenAI GPT models
│   │   ├── googleai.js                 # Google Gemini
│   │   ├── anthropicai.js             # Anthropic Claude
│   │   ├── xai.js                      # xAI Grok
│   │   ├── deepseekai.js              # DeepSeek (OpenRouter)
│   │   ├── groq.js                    # Groq
│   │   ├── chatminds.js               # LM Studio local
│   │   └── pdfqa.js                   # PDF Q&A assistant
│   ├── components/
│   │   ├── Assistant/                 # Model selector dropdown
│   │   ├── Chat/                     # Main chat container
│   │   ├── Controls/                 # Input bar, voice, upload
│   │   ├── Messages/                 # Message bubbles, copy
│   │   ├── Sidebar/                  # Chat list, mobile menu
│   │   ├── SystemInstruction/         # Persona editor
│   │   ├── Theme/                    # Dark/light toggle
│   │   ├── Loader/                   # Loading spinner
│   │   └── Auth/                     # Protected route
│   ├── pages/
│   │   ├── Login.jsx                 # Login page
│   │   └── Registration.jsx           # Registration page
│   ├── utils/
│   │   ├── auth.js                   # Demo authentication
│   │   └── exportPdf.js               # PDF export utility
│   ├── App.jsx                       # Main app & routing
│   ├── App.module.css                 # App styles
│   ├── index.css                     # Global styles
│   └── main.jsx                       # Entry point
├── server/                             # Backend server (optional)
│   └── index.js                       # Express server for vectors/uploads
├── .env.example                       # Environment template
├── package.json
├── vite.config.js
└── README.md

```

Routes

Route	Description	Auth Required
/	Home (Chat UI)	☒ Yes
/Login	Login page	☒ No
/Registration	Registration page	☒ No

Adding a New AI Provider

1. Create `src/assistants/<provider>.js`:

```

export class Assistant {
  constructor(systemPrompt) {
    this.systemPrompt = systemPrompt;
  }

  async chat(content, history) {
    // Return complete response string
  }

  async *chatStream(content, history) {
    // Yield chunks for streaming
  }
}

```

2. Add to `src/components/Assistant/Assistant.jsx`:

- Import the adapter
- Add to `assistantMap`
- Add `<option>` in the selector

Security Notes

 **Development Only:** API keys are exposed in the browser bundle. For production:

- Use a backend proxy to hide API keys
- Implement proper rate limiting
- Replace demo auth with Firebase/Auth0/custom backend

Troubleshooting

Issue	Solution
Provider auth error	Check API key in <code>.env</code> , restart dev server
No voice input	Use Chrome/Edge (Safari limited support)
LM Studio not connecting	Ensure LM Studio server is running on port 1234
Mobile mic not working	Grant microphone permission, use HTTPS in production
PDF export empty	Send at least one message first

Roadmap

- ☐ Backend API proxy for production
- ☐ Firebase/Auth0 integration
- ☐ Conversation search
- ☐ Chat export/import (JSON)
- ☐ Token usage tracking
- ☐ Custom theme colors
- ☐ Multi-user support

Contributing

Contributions welcome! Feel free to:

- Report bugs
- Suggest features
- Submit pull requests

License

MIT License — feel free to use and modify.

Credits

Built with  using React, Vite, and multiple AI SDKs.

ChatMinds — Your unified AI conversation companion.